ENGINEERING DRAWING WITH AUTOCAD

UNIT 5: ISOMETRIC VIEWS & 3D MODELING FUNDAMENTALS

Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Engineering Drawing with an Introduction to AutoCAD by Dhananjay Jolhe

>> 5.1 Principles and Terminology of Axonometric Projections

Axonometric projection is a parallel pictorial projection where the object is tilted so that all three principal axes are visible in a single drawing:
* Types of Axonometric Projections: Isometric (all 3 axes equally inclined at 120 deg), Dimetric (2 axes equally inclined), Trimetric (all 3 axes unequally inclined).
* Isometric Axes: Three lines meeting at 120 degrees to each other. In 2D drawing, one axis is vertical, and the other two axes are inclined at 30 degrees to the horizontal.
* Isometric Lines: Lines parallel to any of the three isometric axes. Distances along isometric lines can be measured directly to scale.
* Non-Isometric Lines: Lines NOT parallel to any isometric axis (e.g., edges of pyramids, oblique chamfers). Cannot be measured directly; must be located via coordinates of their endpoints.
* Isometric Planes (Isoplanes): Three coordinate planes defined by pairs of isometric axes: Isoplane Left (30 deg left & vertical), Isoplane Right (30 deg right & vertical), Isoplane Top (30 deg left & 30 deg right).

>> 5.2 Isometric Projection vs Isometric View

* Isometric Projection: Drawn using foreshortened Isometric Scale. Length = True Length x cos(45) / cos(30) = sqrt(2)/sqrt(3) approx 0.816 (81.6% of True Length).
* Isometric View (or Isometric Drawing): Drawn using True Scale (100% full scale dimensions). The drawing is identical in proportion to isometric projection but is approximately 22.5% larger in visual area.
* Isometric Scale Construction: Draw horizontal line. From origin, draw line at 30 degrees (Isometric scale) and line at 45 degrees (True scale). Project true divisions vertically down onto the 30-degree line.

>> 5.3 Methods of Drawing Isometric Projections

* 1. Box Method: Enclose the entire object inside an imaginary isometric rectangular box of length L, width W, and height H. Carve out features by measuring along isometric box edges. Essential for irregular objects, prisms, pyramids.
* 2. Coordinate / Offset Method: Locate individual key corner vertices in space using X, Y, Z coordinates measured from a defined reference corner.
* 3. Four-Center Method for Isometric Circles (Ellipses): A circle in orthographic view appears as an ellipse in isometric view. Inscribed in an isometric rhombus (rhombus side = circle diameter). Drawn using four arcs struck from the two obtuse angle vertices and the intersections of opposite side perpendicular bisectors.

>> 5.4 AutoCAD 3D Modeling Fundamentals

* 3D Coordinate Systems: World Coordinate System (WCS) is fixed. User Coordinate System (UCS) can be dynamically rotated and translated to align with any 3D face using UCS -> 3Point (Origin, X-axis, Y-axis).
* 3D Primitive Solids: BOX, CYLINDER, CONE, SPHERE, TORUS, WEDGE, PYRAMID.
* 3D Solid Generation Commands:
- EXTRUDE (EXT): Projects an enclosed planar 2D curve or region along a perpendicular height or specified path.
- REVOLVE (REV): Rotates a planar 2D profile around a specified axis of revolution by up to 360 degrees.
- PRESSPULL: Dynamically extrudes an enclosed boundary area or face directly without needing a prior Region entity.
- SWEEP & LOFT: Generates solids along arbitrary 3D guide curves or between multiple distinct cross-sectional profiles.

[PROBLEM 1: Isometric View of a Hexagonal Prism using Box Method]

Problem Statement: A hexagonal prism has base side 30 mm and axis height 65 mm. The prism rests with its base on HP with one base edge parallel to VP. Draw its Isometric View.

Step-by-Step Construction Procedure:

* Step 1: Orthographic Base & Enclosing Box:
- Draw a regular hexagon of side 30 mm in orthographic plan view.
- Enclose the hexagon in a tight rectangle ABCD. Length L = distance across corners = 2 x 30 = 60 mm. Width W = distance across flats = 2 x 30 x cos(30) = 51.96 mm.
* Step 2: Isometric Enclosing Box:
- Draw isometric axes (one vertical, two at 30 deg to horizontal).

- Construct bottom isometric box base of size 60 mm x 51.96 mm on the 30-deg isometric lines.
- Erect vertical lines of height 65 mm from each corner to form the top isometric box face.

* Step 3: Plot Hexagonal Vertices:
- Transfer the coordinate offsets of the 6 hexagonal vertices from the orthographic rectangle edges onto the corresponding edges of the isometric box base and top face.

* Step 4: Final Isometric Outlines:
- Connect the 6 vertices of the top face with solid lines.
- Connect vertical prism edges: the 3 front-facing vertical edges are solid visible lines; rear edges hidden behind the prism body are omitted.
- Connect the 3 visible front base edges.

[PROBLEM 2: Four-Center Method Construction of Isometric Cylinder]

Problem Statement: Draw the Isometric View of a vertical right circular cylinder of base diameter 50 mm and height 70 mm.

Step-by-Step Four-Center Method Execution:

* 1. Top Face Rhombus: Draw isometric rhombus ABCD of side 50 mm on Isoplane Top (axes at 30 deg to left and 30 deg to right). Obtuse angle vertices are at B (top) and D (bottom).

* 2. Finding Centers:
- Find midpoints 1, 2, 3, 4 of rhombus sides AB, BC, CD, DA.
- From obtuse vertex B, draw lines to opposite midpoints 3 and 4.
- From obtuse vertex D, draw lines to opposite midpoints 1 and 2.
- The intersections of these lines establish two sharp centers C1 and C2.

* 3. Striking Ellipse Arcs:
- Center B, radius B-3: Draw large arc 3-4.
- Center D, radius D-1: Draw large arc 1-2.
- Center C1, radius C1-1: Draw small arc 1-4.
- Center C2, radius C2-2: Draw small arc 2-3. (Completes smooth 4-arc top ellipse).

* 4. Bottom Base Ellipse: Project centers B, D, C1, C2 vertically downward by 70 mm. Strike identical lower arcs for the front half 1-2-3.

* 5. Tangent Generators: Draw two vertical lines tangent to the outer extremities of the top and bottom ellipses. Erase construction guides.

[PROBLEM 3: Composite Solid: Sphere Resting Centrally on a Cube]

Problem Statement: A solid sphere of diameter 40 mm rests centrally on the top face of a cubical block of side 60 mm. Draw the Isometric Projection of the composite assembly.
Mathematical Derivation & Construction Protocol:
* 1. Cube Construction (Isometric Scale): Side of cube in isometric projection = 60 mm x 0.816 = 48.96 mm. Draw isometric box of side 49 mm.
* 2. Locate Center of Sphere:
- Find center of top face of cube by intersecting diagonal lines.
- The geometric center of the sphere lies vertically above the top face at a distance equal to the sphere radius.
- Isometric height of center = 20 mm x 0.816 = 16.32 mm above the cube face. Mark sphere center point O.
* 3. Sphere Outline in Isometric Projection:
- CRITICAL RULE: In axonometric / isometric projection, a sphere always projects as a TRUE CIRCLE! It does NOT become an ellipse.
- Radius of circle in projection = TRUE radius of sphere = 20.0 mm (NOT foreshortened, because the sphere contour in all directions remains perpendicular to sight rays).
- Center at point O, draw a true circle of radius 20 mm.
* 4. Visible Outlines: Top face of cube partially hidden beneath sphere silhouette; finish visible edges with solid thick lines.

[PROBLEM 4: AutoCAD 3D Solid Modeling Sequence for Stepped Flange]

Problem Context: Model a 3D stepped shaft with circular flange and counterbored hole using AutoCAD 3D tools.
AutoCAD Command Protocol:
* 1. Workspace: Switch to "3D Modeling" workspace. Set Visual Style to "Shaded with edges".
* 2. REVOLVE Method:
- Draw 2D half-cross-sectional profile using PLINE: Start at 0,0 -> 15,0 -> 15,60 -> 45,60 -> 45,80 -> 0,80 -> Close.
- REVOLVE -> Select polyline -> Axis of revolution: 0,0 to 0,80 -> Angle: 360 deg. (Generates stepped solid cylinder).
* 3. CYLINDER for Central Bore:
- CYLINDER -> Center: 0,0,0 -> Radius: 15 mm -> Height: 80 mm.
* 4. SUBTRACT Command:
- SUBTRACT -> Select outer stepped solid -> Press Enter -> Select inner cylinder -> Press Enter. (Produces hollow through bore).
* 5. 3P UCS for Flange Bolt Holes:
- UCS -> 3P -> Click flange center, X-axis, Y-axis.
- CYLINDER -> Center: 30,0,60 -> Radius: 4 mm -> Height: -20 mm.

- ARRAYPOLAR -> Select small cylinder -> Center 0,0 -> Items: 6 -> Angle: 360.
- EXPLODE array -> SUBTRACT the 6 cylinders from main flange body.